## PROBLEM STATEMENT PARTNERS

# ROUND 2 - PROBLEM STATEMENTS

| PS ID | Title | Description |
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| R2-01 | Secure Collaborative Workspaces in the Age of BYOD | Employees in modern workplaces increasingly access enterprise systems through personal devices under BYOD (Bring Your Own Device) policies, creating serious security and privacy challenges. Propose a cybersecurity solution that enables secure, auditable access to organizational resources from unmanaged devices while preserving employee privacy. The system should follow zero-trust principles, support granular access control, and ensure real-time monitoring without compromising usability or compliance. |
| R2-02 | Secure Cross-Device Clipboard for Remote Teams | As remote and hybrid work environments become standard, professionals regularly switch between devices—copying sensitive data such as passwords, code snippets, and confidential notes. Existing clipboard syncing tools offer convenience but fall short on security, visibility, and organizational control. Your challenge is to develop a secure, cross-device clipboard synchronization system tailored for distributed teams. The solution must include end-to-end encryption, domain-specific access controls to block data from being pasted into untrusted apps or websites, and a tamper-proof log of all transfers. Prioritize compliance, privacy, and usability to meet the needs of developers, regulated industries, and security-focused organizations. |

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| R2-03 | Cross-Border Medical Records via Interoperable Blockchains | Patients often face difficulties accessing their medical records while traveling internationally due to fragmented systems and privacy regulations. Propose a blockchain-based solution that enables secure, cross-border sharing of medical data with role-based access control, patient-managed consent, and interoperability across multiple blockchain networks like IPFS, Hyperledger, and Ethereum. The solution should prioritize privacy, compliance, and real-world usability in global healthcare scenarios. |
| R2-04 | AI Companion Quest – Smart Gamified Learning Buddy | Learners on video-based platforms often face boredom, distraction, and a lack of emotional connection, leading to poor engagement and retention. Traditional platforms provide static experiences that don't adapt to individual learning styles, pace, or mood. Your challenge is to build an AI-powered learning companion that personalizes the experience by responding to user emotions, interaction patterns, and progress. The system should include gamified features like streaks, quests, and point boosters; adjust its tone and personality based on emotional cues; and recommend relevant hands-on or DIY activities post-lessons to reinforce learning and maintain engagement. |
| R2-05 | DIY Mission Engine – Auto-Generate Build Tasks with AI | Students often feel motivated to build something after learning a new concept but struggle to decide what to create or where to start, leading to lost opportunities for hands-on exploration. Your challenge is to develop an AI-powered DIY project generator that takes input such as a concept name or video lecture transcript and generates personalized build tasks. The system should adapt task complexity to the learner's background, suggest project types across domains like coding, hardware, design, or research, and offer templates or guided hints using large language models to help learners begin with confidence. |

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| R2-06 | AI Gamifier – Convert Courses into Interactive Game-Based Modules | Static course content often fails to engage college students, especially when compared to the immersive, rewarding nature of games. Most educational platforms lack built-in game design logic that could enhance motivation and retention. Your challenge is to build an AI-powered gamification engine that transforms traditional video lessons or text modules into interactive learning experiences. The system should generate mission-style learning paths, level-up systems tied to learning outcomes, and game-like interfaces featuring rewards, progress tracking, and achievement mechanics to make studying feel more like playing. |
| R2-07 | Sentiment-Aware AI Coach – Real-Time Feedback Based on Emotion & Confusion Detection | In large-scale online courses, instructors can't provide real-time support, leaving many students feeling confused, frustrated, or disengaged—limiting the impact of self-paced digital learning. Your challenge is to build an AI-powered feedback coach that detects emotional signals like confusion, boredom, or frustration using webcam, audio, and interaction data (e.g., mouse movement, inactivity). Based on these cues, the system should suggest timely resources such as videos, mini-games, chatbots, or breaks. It should also track emotional well-being alongside cognitive progress to create a more personalized and responsive learning environment. |
| R2-08 | Immersive Pollution Exposure Simulator for Policy Impact & Public Awareness | Despite worsening pollution in Indian cities, there's a disconnect between environmental data, public awareness, and effective policy-making. Citizens lack clear insight into how pollution impacts their daily lives, while policymakers struggle to act on complex data. Your challenge is to develop an AR/VR-based interactive platform that visualizes real-time exposure to air, water, and noise pollution. The system should integrate real or simulated datasets, use AI to simulate long-term human impacts (e.g., health risks from PM2.5 or waterborne diseases), and enable “What if” urban planning scenarios. It should feature tailored modes for citizens, schools, and policymakers, using goal-based storytelling and decision-driven simulations to drive education, engagement, and action. |

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| R2-09 | Mixed-Reality Urban Heat Island Visualizer for Sustainable Urban Planning | Urban heat islands (UHIs), driven by concrete buildup, deforestation, and poor urban design, raise local temperatures, increase health risks, and strain energy resources—yet their effects remain largely invisible to citizens and planners. Your challenge is to build an AR-based spatial visualizer that overlays UHI data—such as temperature gradients, green cover loss, and surface reflectivity—onto real-world cityscapes using mobile AR or mixed-reality devices like HoloLens. The system should pull from real-time IoT or satellite data, simulate interventions like green roofs or reflective pavements, and support on-site use by planners, architects, and citizens. Optionally, include AI-driven suggestions for cool-surface retrofits and green buffer strategies to help mitigate UHI effects. |

*Equipments will not be provided

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